

Institute of Actuaries of India

Subject ST1 – Health and Care Insurance Specialist Technical

October 2015 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**(i)**

- New business is analysed primarily to calculate capital strain and profitability.
- The company needs to be sure that it has sufficient capital to finance the new business strain caused by the policies it sells.
- So it will need to monitor the volume of new business sales over time against targets regularly.
- ... because the higher the volume of sales in any period, the more capital will be needed.
- It will also need to analyse the new business mix ...
- ... because some policy types will require more capital than others.
- The volume of sales needs to be monitored so that the company can check that its per-policy expense assumptions remain correct.
- This is because the company's overheads are shared between all policies sold by the company, and the per-policy expense assumption is therefore a function of the assumed volume of future sales.
- As overheads are mostly staff costs, then falling sales may create a problem because the company will end up with excessive staffing capacity.
- Conversely, too much new business may lead to inadequate capability to deal with the work required like underwriting, processing the proposal form etc.
- ... leading to processing errors, service delays and the consequent deterioration of public image
- New business profitability is also requires to be analysed on regular basis. The purpose is to check the variance in the profitability at the time of pricing versus when new business is actually sold.
- Then reasons for variances should be analysed. If it is due to some experience changes, New Business mix, assumption change, change in payment to distribution channel etc.
- Other analysis includes IRR and Discounted Payback Period of new business sold.

[7]**(ii)****Significance of the persistency experience analysis**

- The actuary will review the lapse and persistency experience of the business written as renewal of the cover under the indemnity based medical products are optional
- Actual lapse rate will be compared with assumed rate for the region and distribution channel etc.
- Such analysis will help in sales management, commission clawback and understanding the marketing impact
- This is significant as the initial costs are amortised over a number of years of renewal and any adverse experience will affect the recovery of initial expenses and hence the profitability

- Such short term products do not make profits until the policy remains with the same company for few years so early lapse leads to value loss.
- This becomes all the more significant when the market is competitive and portability of policies is allowed
- Therefore such analysis helps insurer to take action to increase persistency level and keep them in line with the assumption or else
-re-price the product if the actual experience turns out to be significantly different from the assumed.

[4]

[11 Marks]

Solution 2:

- First of all, check if all the benefits are modelled in the pricing model
- Check if the minimum and maximum term, premium, sum assured are taken in account while selecting the model points.
- Ensure that benefits are as per regulatory restriction
- Check the proportion of surrender values offered in the product. Is it fair to customer.?
- What is the strain on profitability if surrenders increase or decrease.
- Check all the assumptions are correctly inputted in the model.
- Ensure tax rates and RSM factors are as per regulations.
- Check that projections within the model are reasonable.
- Check that lapse rates have been correctly applied within the cashflow projections, i.e. that any year dependent or duration dependent rates apply to the correct year or duration.
- Check that expense output from the model increases in line with the inflation assumption.
- Check that premiums and sums insured increase each year in line with the relevant increase option (i.e. no increase, RPI, or fixed percentage).
- Check that no claims outgo is paid during the deferred period.

Output checks:

- Carry out spot checks on some calculations to ensure correct.
- Check that the profitability indicated by the model is reasonable.
- For all model points, check the consistency in profitability. Example: for same model points check the profit margin for a policy sold to male or a female.
- Profitability can be compared to output from previous pricing exercises and to new business profitability information produced by a reporting team.
- Check that profitability rates are in line with profitability requirements that have been set.
- Could check against the results from a formula approach.
- Check that the sensitivity analyses give sensible results.

Premium rates:

- Check that premium rates are consistent with the objectives set for the pricing exercise; for example, it could be to reduce premium rates to be more competitive or to increase rates to improve profitability or it might have been due to changing the benefit definition/term.
- Check that premium rates look sensible by “model point” e.g. higher premium rates for older lives, but need to consider that premiums for ages nearing the ceasing age would reduce (due to the potential benefit term reducing); higher premium rates for shorter deferred periods compared to longer deferred periods; higher premium rates for occupation classes that pose a higher risk; higher premium rates for increasing cover compared to level cover.
- Compare results against competitor rates.

[10 Marks]**Solution 3:****(i) Appropriate reinsurance arrangement**

- a. The reimbursement based individual policies for individuals as well as the Group IP policies can experience very large claims cost which cannot be capped under a proportional reinsurance arrangement
- b. Under excess of loss reinsurance (XL) , non-proportional insurance arrangement on the other hand the cost to an insurer of a large claim from an individual policy or under a group cover is capped with liability above a certain level often called as lower limit or retention and is therefore appropriate to the nature of the policies being sold by the health insurer
- c. However in case the claim amount exceeds the upper limit of reinsurance the excess will revert to the insurer.
- d. Variations of this form of reinsurance cover exist to limit an insurer’s loss from a single event or over a given period.
- e. Following key elements would apply in use of a non- proportional reinsurance arrangement
 - Application of a lower limit on individual claims to cover against large individual risks and possibly on aggregation of claims to cover against claims coming under claim from Group IP
 - Decision on the upper limit above which reinsurer’s liability ceases
 - The reinsurer might pay for all claims within the limit or proportion of claims
 - Limits might be linked to inflation or defined escalation rates
- f. Under this arrangement of XL the reinsurer will indemnify the insurer for amount of any loss above a state excess point but upto a stated upper limit
- g. The insurer can purchase further layers of XL from different reinsurers which stack on top of primary layer.
- h. While the primary layer has an upper limit the top layer may be unlimited or as is decided
- i. The layers shall generally be arranged so that there is no gap
- j. In view of the significant impact of inflation a stability clause applied to excess point provides for indexation of monetary limit in line with specified inflation index. The upper limit may similarly be indexed to preserve the original real value to the cover.

- k. Inflation Indexation helps in more equitable division between insurer and reinsurer. In event of excess point not being indexed the insurer would pay extra premium for the added risk to reinsurer
- l. Return commission and override commission may not be relevant but profit commission might be provided
- m. The XL may be risk excess of loss type that relates to individual losses which are large above the excess point
- n. Aggregate XL or stop loss might apply when the excess point and upper limit apply to aggregation of multiple claims rather than on the large individual claim
- o. Non proportional insurance is unlikely in long term health insurance but is appropriate for the aggregate excess of loss for group IP covers
- p. The excess point and upper limit for the aggregate XL may be defined in terms of percentage of premium income under that account

[6]

(ii)

Advantages of non-proportional reinsurance

- Permits acceptance of risks that could lead to large claims as compared to annual premium and free assets of insurer and where the claim amount is not defined in terms of sum assured
- Reduces risk or insolvency from aggregation of claims due to single cause
- Stabilises technical results by reducing claim fluctuation
- Reduces volatility in annual profitability
- This enhances more efficient use of capital
- Company can write more business for a given amount of capital

Disadvantages of Non Proportional insurance

- Cost in form of reinsurance premium to be paid
- This in long term might be more than expected recoveries under treaty
- The cost of reinsurance would be expected to be higher as the reinsurer will load claims for profits besides the expected cost of expenses
- At times XL Premiums may be considerably greater than the pure risk premium for the cover
- This may happen particularly after insurer having had a few years of poor claims experience

[4]

[10 Marks]

Solution 4:

- (i) Embedded value of a health insurance company is defined as the sum of the market value of net assets and the present value of future profits net of tax and cost of capital from existing business, discounted at the risk discount rate.

[2]

(ii) Analysis of movements in the embedded value allows insurers to----

- Reconcile the values between successive years
- Validate the calculations assumptions and data used
- Provide data for use in executive remuneration schemes
- Provide management information
- Provide detailed information for publication in the accounts of the company or of parent company also for the new business taken on by the company
- Identification of lines of business based on profitability
- Assess the yearly return on capital
- Review and reassess the views on future with reference to assumptions or models used for pricing and reserving
- Its review would also help in reviewing the way in which business and risks are managed

[2]**(iii) Statutory reserves of a health insurance company selling long term individual health insurance products are calculated based on following set of broad principles**

- The nature of liabilities is long term and hence amount of reserves should be so as to ensure that all liabilities can be met in long term
- Liabilities should be calculated by prudent valuation of all future liabilities for all existing policies including guaranteed benefits, options , expenses taking credit for premium that are due to be paid
- Since the term of the products is quite long, valuation should include appropriate margins for adverse deviation
- The valuation should take into account the nature, term and methods of valuation of corresponding assets for these type of policies
- Use of generalisation or approximation should be allowed for
- For the long term nature of these products, rate of interest for calculation of reserves should be taken prudently and having regards to yields on corresponding existing assets
- The elements of statistical basis that is demographic and persistency assumptions and allowance for expenses for calculating reserves should be taken prudently
- Amount so defined in advance for taking into valuation should not be such as to be less than prudent estimate of relevant future expenses
- Method of calculation of reserves from year to year should be such as to recognise profit in an appropriate way over duration of policy
- Each insurance company to disclose the methods and bases used in valuation

[5]**(iv) Following Reserves for long term health insurance Products will need to be maintained**

- The discounted vale of future expected claims expenses and premium cash flows
- For claims that have been incurred but not reported (IBNR)

- For claims that have been reported and not yet fully settled
- Reserves for additional costs for options that would come into money

[2]

(v)

- The basic purpose of supervisory reserves is to ensure that reserves are set in prudently
- Prudency requires supervisory reserves to be set in a manner so that in majority of cases reserves are higher than the surrender value of each policy
- At the start of the policy the surrender values are expected to be higher than the asset share at times due to marketing reasons or due to regulatory stipulations. In such a case the supervisory reserves can be set subject to a minimum of any surrender value.
- The company should be holding enough reserves to meet the call on money if many people decide to surrender simultaneously.
- If the company holds reserves subject to a minimum of any surrender value then even if policies surrender at once any time in future, the company will have more than sufficient reserves to pay them and still make profit
- The supervisory reserve that ignores the surrenders in future will be more than the one calculated taking in to account the future surrenders. Therefore if supervisory reserve ignores future surrenders then when a policy is surrendered a profit will still be made equal to excess of supervisory reserve at any time over surrender value paid at that time.
- However, in case of provision of guaranteed surrender value being higher than supervisory reserves at times of surrender, then future surrender can cause loss and a larger reserve will be required to cover them.
- Here again the policies be valued on assumption that all of them will surrender at the worst possible time from company's perspective.

[4]

[15 Marks]

Solution 5:

(i)

- Deferred period
- Waiting Period
- Linked claim period
- Replacement Ratio
- Benefit escalation rate – fixed or variable
- Minimum and Maximum benefit amount
- Death benefit
- Surrender benefit
- Paid up benefit
- Waiver of premium
- Options and guarantee

- Rehabilitation facilities
- Expiry age/term
- Premium Frequency
- Definitions of the contingent events
- Exclusions

[4]

(ii) The rating factors would be:

- Age
- Smoker/Non-smoker
- Location
- Years of experience
- Deferred period
- Linked claim period
- Distributional channel
- Type of gym instruction carried out
- Type of sports taught
- Hours per week worked
- History of work-related injuries
- Whether any martial arts are taught

[3]

(iii) The product is simple and low cost and therefore less expensive distribution channel would be most appropriate.

Target market is very specific and hence distribution channel should concentrate on that population only.

Inserts in Gym or health related magazines are appropriate way to advertise the products.

Gym instructor website advertising is also appropriate.

Online channel can be used using own website to market the product.

Also company can sign up with professional bodies to buy their mailing lists to target the specific population.

[3]

(iv)

- The biggest risk is the morbidity risk. Gym instructors are expected to be fit but sometimes small injuries prevent them to return to work.
- The frequency of claim and length of claim is very uncertain. Hence, there is a risk of mis-estimation of claims assumptions while pricing the product.

- There is a risk of non-disclosure of pre-existing conditions/injuries.
- Anti-selection risk – those people who thought they were more likely to claim would naturally be most likely to take out the product and these may be difficult to identify from the limited underwriting.
- The definition of work-related injuries may not be clear, and may be open to moral hazard.
- The exact definition of unable to work will be defined, but this may not be examined fully at time of purchase and therefore risk of legal interpretation.
- If there are cross-subsidies, this introduces mix risk – in particular, some sports may be more hazardous than others but it may not be possible or practical to adequately reflect the variation in the rating factors, even if the data were available.
- Competitors may issue a similar product and undercut the company.
- The premium may be high relative to the benefit payable. This introduces both volume risk and risk of high non-renewal rates of good risks. Hence, the Company will left with only bad risks.
- Risk of not selling sufficient volumes to recover the development overheads.
- Potential for fraud.
- Expense Risk: Actual expenses are more than expected

[6]

[16 Marks]

Solution 6:**(i) Principles of Investment Management for a health insurance company**

- As a basic principle health insurer should select investments in a manner appropriate to the nature, term and currency of liabilities.
- The investments should also be selected so as to maximise the overall return on the assets where overall returns includes both investment income and capital gains.
- The extent to which matching can be departed from in order to maximise the overall returns will depend upon the extent of the free assets of the company and the risk appetite of the company.

[3]

(ii) Comments on the suggestion by the Board member

- Private medical insurance is a short tailed business with very little scope for significant amounts being available for investments and there by generating investment returns.
- The insurer selling the PMI policies would be required to monitor costs arising from various treatments that are covered under PMI to determine any amounts that would still be available for investments

- It is hoped that at least approximately the premium income from all policies should be sufficient to pay the claims and to cover the company's expenses.
- Therefore the reserves under PMI policies are very small and hence the investment income is also very small
- While from the long term health products, much larger amounts will be available for investments and generation of investment returns. Therefore the investment returns under short term product like PMI cannot be compared with long term Health Insurance Products.
- However assets will still be required and hence a mix of cash and short dated fixed securities attuned to the nature of these liabilities are found suitable by fund managers.
- Investing long term of assets for PMI , which are not matching to the nature of the liabilities would be risky due to asset liability mismatch
- However where the insurer is aware of the future treatment that could persist over some future years or due to legal delays in settlement of claims etc. the insurer would still establish reserves.
- The exact amounts to be set up might not be known at the time of setting such reserves and these would further escalate in line with medical inflation often higher than price inflation upto the time of final settlement or legal award
- In such cases assets can be recommended which can give real return over the relatively short period until the case goes off the books
- To match the needs where the real returns over the short period are expected investments could be considered in index linked bonds.
- However the claims speed could still be higher than of these bonds or the availability of such bonds by itself may be a problem
- In such cases the investments can then be considered in long term instruments such as equities or properties
- In such circumstances the volatility of such assets particularly over shorter periods and their mismatch with the liabilities has to be considered
- Hence the fund manager for the short duration PMI product cannot be suggested to toe the line of investment pattern for long term business in view of the difference in duration of the portfolio and small amount of reserves under PMI Products.
- Overall the assets of the shorter term Products like PMI would therefore expected to earn lesser returns as compared to the longer term health insurance Products.

[5]

[8 Marks]

Solution 7:

There are many risks in adding the new annually reviewable reimbursement based product, for which the insurer does not have any previous experience as its existing product portfolio consists of fixed benefit based product only. Some of risks are as under:

- Insurer may suffer worse claim experience than allowed in pricing due to higher claim incidence and higher claims amounts
- Underwriting procedure being unable to cope up with assumptions made in the pricing basis according to nature of this product.
- Lack of proper disclosure of information by buyers and inability to find out the suppression of information at claims stage coupled with inapt claims control and monitoring capacity of the company.
- Moral hazard causing higher than expected claims experience
- In-appropriate reinsurance policy
- Inability of the insurer in wording the contracts to rule out claims not indented to be paid.
- Lack of expertise in resources and systems to control and monitor claims and risks
- Lack of credible data and inability in assimilation of experience data for future use
- Lack of appreciation on the part of management to understand the different nature of this business and the type of expert resources such as medical doctors required to control claim costs for example the approach of doctors and underwriter itself may need to be attuned to this class of business
- Inability to control expenses of management and distribution in the product
- Mis-selling by the distribution channel may **result in higher lapse rates.**
- Frauds or lack of management by the third parties like TPAs.
- Anti-selection by policyholders and inability of insurer to control the same can work to the disadvantage of the insurer.
- Selective renewals by higher claiming lives and good lives withdrawing.
- Inability OF insurer to claw back high commissions paid in early part for the lapsing policies.
- Inability to sell sufficient volumes due to competition or lack of proper marketing strategy.
- Lower policy renewal due to higher premiums than competitors or lack of customer servicing.
- Inadequacy of capital to support large volume of sales thereby restricting the sales. However this is less likely to be the case as the company overall is large and should be able to provide the capital.
- Change in volume and mix of business from the from the assumption made in pricing. This may happen due to sales to a different target market where the claim experience is worse.
- Sales of larger number of smaller ticket size policies than expected on average would put strain on the per policy expenses.
- In order to be competitive and increase market share, if the premiums are kept artificially low, it may lead to losses. The expectations created may make reprising at the time of review difficult which may be further subject to prior approval of regulator

- Features such as guarantees, waiver of waiting periods for competitive reasons may make the product loss making
- Procedural relaxation such as relaxed underwriting etc may encourage anti-selection and more losses rather than compensating through increase in sales volumes.
- The inappropriate activities of marketing channels and other third parties may bring reputation risk to the business
- Changes in tax and related regulations may make this product unattractive to buyers and/ or unprofitable to insurers. For example removal of tax incentives may make the product less attractive and reduce number of sales. Compulsory renewability if mandated by regulation may put pressure on profit margins
- The company may suffer from asset liability mismatch. For example the company may face liquidity problems if assets are invested longer than liabilities causing liquidity problems.

[10 Marks]

Solution 8:**(i)**

1. Protecting the nation's health
 - safeguard a healthy and productive workforce
 - promotes productivity and a growing Gross Domestic Product (GDP)
2. Redistribution of wealth
 - Ensure that the poorest members of society have access to medical care
 - Ensure adequate provision of services to the elderly and for children
3. Balancing the budget
 - The cost of provision can be funded through specific health charges or via general taxation, or a mixture of the two.
 - Higher earners pay more
4. Following social culture and/or political promises.
 - State healthcare programmes may be part of national culture
 - Government may have made electoral promises

[4]

(ii)

- First of all, what should be the features of the State healthcare provisions?
 - ✓ Cash benefits or care only
 - ✓ Length of benefits paid e.g. 2 years
 - ✓ Death benefit, if any
 - ✓ Maximum per day cost in case of hospitalization
 - ✓ Maximum total benefits
 - ✓ Fixed or regular incomes
 - ✓ a salary-related benefit or fixed benefit
 - ✓ Triggering conditions for the claim
 - ✓ Deferred period – 3 month or 6 month

- ✓ Benefit escalation rate
- ✓ Long term care benefits, if any
- This largely depends on population profile of the country.
- And varies for different lifestyles and earning capacity of the population.
- Level of cross-subsidies between rich and poor, healthy and unhealthy.
- Mean testing benefits to target benefits to those most in need. The State may provide little or no benefit to individuals with sufficient private means, and instead direct resources to the less well-off to ensure that everyone achieves a certain minimum standard of living when in ill-health.
- Funding method to be chosen: pay as you go or forward funding or a combination of two.
- The benefits may be provided by its own medical establishments or provided by commercial establishments (that are subsequently reimbursed).
- Since the purpose of the scheme is to reduce funding burden on government. Hence, government may offer incentives for self-provision of healthcare benefits, such as:
 - ✓ tax relief on healthcare insurance premiums
 - ✓ exclusions from State benefits
 - ✓ reduced general taxes where insurance is taken out
 - ✓ subsidies to private purchase of healthcare services.

[10]

(iii)

Advantages:

- The initial benefit is at such a level that there will not be an abrupt change in lifestyle as a result of sickness.
- It is less likely that benefits (in the first 26 weeks) will exceed normal earnings, and so an incentive to return to work remains.
- Post 26 weeks, linked to average earnings, so the claimant will benefit from the economic growth of the country.
- Post 26 weeks: May only be sufficient to cover basic living costs, so, unless pre-sickness income was lower than the sickness benefit, it will act to encourage a return to work as soon as possible.
- Post 26 weeks, administration will be easy and less costly.

Disadvantages:

- Complicated to understand
- It may take time to collect all the relevant information.
- The administrative cost will be greater.
- The initial incentive does not encourage returning to work as this is quite high.
- There is no waiting period.
- Overall the benefits are likely to have a higher cost.

- Post 26 weeks, the benefit may be higher than they would earn when able to, which would be a significant disincentive for the person to return to work, and would increase the cost of providing the benefits as a result of the moral hazard.
- Post 26 weeks, the benefit decreases which may still be higher than they were earning when able to, which would be a significant disincentive for the person to return to work, and would increase the cost of providing the benefits as a result of the moral hazard.

[6]

[20 Marks]
